

Very-low-carbohydrate ketogenic diet v. low-fat diet for long-term weight loss: a meta-analysis of randomised controlled trials.

The role of very-low-carbohydrate ketogenic diets (VLCKD) in the long-term management of obesity is not well established. The present meta-analysis aimed to investigate whether individuals assigned to a VLCKD (i.e. a diet with no more than 50 g carbohydrates/d) achieve better long-term body weight and cardiovascular risk factor management when compared with individuals assigned to a conventional low-fat diet (LFD; i.e. a restricted-energy diet with less than 30% of energy from fat). Through August 2012, MEDLINE, CENTRAL, ScienceDirect, Scopus, LILACS, SciELO, ClinicalTrials.gov and grey literature databases were searched, using no date or language restrictions, for randomised controlled trials that assigned adults to a VLCKD or a LFD, with 12 months or more of follow-up. The primary outcome was bodyweight. The secondary outcomes were TAG, HDL-cholesterol (HDL-C), LDL-cholesterol (LDL-C), systolic and diastolic blood pressure, glucose, insulin, HbA1c and C-reactive protein levels. A total of thirteen studies met the inclusion/exclusion criteria. In the overall analysis, five outcomes revealed significant results. Individuals assigned to a VLCKD showed decreased body weight (weighted mean difference 20.91 (95% CI 21.65, 20.17) kg, 1415 patients), TAG (weighted mean difference 20.18 (95% CI 20.27, 20.08) mmol/l, 1258 patients) and diastolic blood pressure (weighted mean difference 21.43 (95% CI 22.49, 20.37) mmHg, 1298 patients) while increased HDL-C (weighted mean difference 0.09 (95% CI 0.06, 0.12) mmol/l, 1257 patients) and LDL-C (weighted mean difference 0.12 (95% CI 0.04, 0.2) mmol/l, 1255 patients). Individuals assigned to a VLCKD achieve a greater weight loss than those assigned to a LFD in the longterm; hence, a VLCKD may be an alternative tool against obesity.